Master Question Bank

Includes difficulty tags and a full answer key with explanations at the end.

Questions

Section A: MCQ (20)

MCQ01 [Easy] Robotics is best described as:

A) Study of computer programming only

B) Integration of mechanics, electronics, and computing

C) Purely artificial intelligence research

D) Mechanical engineering without control

MCQ02 [Easy] The Sense–Plan–Act model represents:

A) Hardware architecture

B) Power distribution

C) Perception → Decision → Action sequence

D) Only sensor processing

MCQ03 [Easy] A reactive paradigm robot:

A) Uses a global world model

B) Depends only on real-time sensor inputs

C) Plans entire path before moving

D) Avoids using sensors

MCQ04 [Easy] The main weakness of hierarchical control is:

A) Too fast reaction

B) Lack of planning

C) Slow response to dynamic changes

D) No goal orientation

MCQ05 [Easy] Hybrid paradigm combines:

A) AI + IoT

B) Planning + Reactive layers

C) Sensors + Actuators

D) Motors + Power

MCQ06 [Easy] Which actuator is best for heavy-duty high-force tasks?

A) Electric

B) Pneumatic

C) Hydraulic

D) Servo

MCQ07 [Easy] Stepper motors are preferred because they:

A) Require hydraulic oil

B) Give precise angular steps

C) Run only on AC

D) Have no rotor

MCQ08 [Easy] A servo motor operates using:

A) Open-loop control

B) Random switching

C) Closed-loop feedback

D) Manual adjustment

MCQ09 [Easy] Which transmission converts rotary motion to linear motion?

A) Worm gear

B) Spur gear

C) Rack and pinion

D) Chain drive

MCQ10 [Easy] Lidar primarily measures:

A) Temperature

B) Distance using laser reflection

C) Magnetic field

D) Air pressure

MCQ11 [Easy] Localization helps robot determine:

A) Battery life

B) Its position

C) Motor torque

D) Network speed

MCQ12 [Easy] The Four D’s help decide:

A) Sensor calibration

B) When to use robots

C) Motor selection

D) Programming language

MCQ13 [Easy] A microcontroller is best suited for:

A) High-level AI

B) Low-level embedded control

C) GPU computation

D) Cloud storage

MCQ14 [Easy] A single-board computer is used for:

A) Simple LED control

B) High-level processing like vision

C) Only power regulation

D) Pure mechanical control

MCQ15 [Easy] Pneumatic actuators use:

A) Electricity

B) Pressurized air

C) Oil

D) Steam

MCQ16 [Medium] Worm gear systems are known for:

A) High efficiency

B) Silent smooth motion with high torque

C) Zero friction

D) Low torque

MCQ17 [Easy] In robotics, “Act” subsystem refers to:

A) Sensor processing

B) Physical motion execution

C) Planning algorithm

D) Communication

MCQ18 [Easy] Asimov's First Law prioritizes:

A) Robot safety

B) Human safety

C) Obedience

D) Automation

MCQ19 [Medium] In reactive systems, memory usage is:

A) Extensive

B) Partial

C) nan

D) Global

MCQ20 [Medium] Hybrid systems are strong because:

A) They remove sensors

B) They freeze less in dynamic environments

C) They avoid planning

D) They avoid feedback

Section B: True/False (20)

TF01 [Easy] Robotics combines sensing, planning, and acting subsystems.

TF02 [Easy] Reactive systems rely on global world models.

TF03 [Easy] Hybrid control integrates planning and reactive layers.

TF04 [Easy] Hydraulic actuators are unsuitable for heavy loads.

TF05 [Medium] Lidar performs well in low-light conditions.

TF06 [Easy] Microcontrollers consume more power than microprocessors.

TF07 [Medium] Stepper motors can hold position without feedback when energized.

TF08 [Easy] The Four D's identify tasks suitable for human labor.

TF09 [Easy] Localization is different from path planning.

TF10 [Easy] Servo motors operate using feedback control.

TF11 [Easy] Pneumatic systems are typically clean and fast.

TF12 [Easy] Hierarchical systems respond instantly to unexpected obstacles.

TF13 [Easy] AUV operates underwater autonomously.

TF14 [Easy] Electric actuators require oil for operation.

TF15 [Medium] In reactive control, behavior is tightly coupled with perception.

TF16 [Easy] Chain drives are quieter than belt drives.

TF17 [Easy] PLCs are commonly used in industrial automation.

TF18 [Easy] Asimov’s laws focus on robot rights over human safety.

TF19 [Easy] Hybrid systems think globally and act locally.

TF20 [Easy] Rack and pinion systems provide linear motion from rotation.

Section C: Descriptive (5)

DES01 [Medium]

A university corridor-cleaning robot must mop long hallways and turn into side corridors. Some floors are glossy black marble, others are reflective glass-like tiles. The robot must avoid false readings from reflections, detect walls and people reliably, and still follow an efficient route.
a) Select the most appropriate control paradigm (Reactive, Hierarchical, Hybrid) and justify.
b) Choose the best primary sensor set (LiDAR, camera, ultrasonic/sonar) for this environment and explain why.
c) Recommend a motor type and a transmission (gear/belt/chain) for the mop arm, focusing on smoothness and noise.

DES02 [Hard]

A factory warehouse robot moves boxed components between stations. It carries small-to-medium loads, drives over slightly uneven concrete, and operates in dusty aisles with occasional low lighting. It must keep stable motion, avoid obstacles (workers, pallets), and remain reliable with minimal maintenance.
a) Choose a suitable actuator type (Electric, Pneumatic, Hydraulic) and justify based on load and maintenance.
b) Select a control paradigm (Reactive, Hierarchical, Hybrid) and justify for a changing warehouse.
c) Pick one obstacle-detection sensor (Ultrasonic, LiDAR, Infrared/Camera) and explain why it fits the dusty setting.

DES03 [Medium]

A classroom-assistive robot must pick up delicate lab items (markers, small beakers, test tubes) from tables and shelves. It must approach slowly, align the gripper carefully, and place items precisely without slipping or collision. Students may move around the robot unexpectedly.
a) Choose the control paradigm (Reactive, Hierarchical, Hybrid) and justify for safe manipulation.
b) Recommend the most suitable motor type (DC, Stepper, Servo) for the gripper/arm joints and explain.
c) Select the best sensing approach (Camera, LiDAR, Sonar/Ultrasonic) to locate objects and guide grasping accurately.

DES04 [Hard]

An outdoor agricultural robot performs row-by-row inspection and light spraying in a field. It faces strong sunlight, shadows from plants, uneven terrain, and occasional loss of accurate GPS signal. It must maintain steady motion, avoid obstacles (tools, irrigation pipes), and keep its row-following behavior consistent.
a) Choose a control paradigm (Reactive, Hierarchical, Hybrid) and justify for outdoor uncertainty.
b) Recommend a practical sensor combination to stay aligned with rows and detect obstacles (e.g., camera, LiDAR, ultrasonic, IMU/GPS).
c) Select an actuator type (Electric, Pneumatic, Hydraulic) for the drive/sprayer mechanism and justify based on field use.

DES05 [Medium]

A high-speed industrial robotic arm performs repetitive pick-and-place on an assembly line. It must achieve high precision, repeatability, and smooth motion for thousands of cycles per day. Downtime is costly, and the system must remain stable under continuous operation.
a) Choose the best motor type (DC, Stepper, Servo) and justify for precision and repeatability.
b) Recommend suitable control hardware (Microcontroller, PLC, SBC) and justify for industrial reliability.
c) Choose a transmission approach (gear/belt/chain) and justify based on stiffness, precision, and maintenance.

Answer Key + Explanations

A) MCQ Answers

MCQ01 [Easy] Answer: **B**

Explanation: Robotics integrates mechanics, electronics, and computing into embodied systems that sense, plan, and act.

MCQ02 [Easy] Answer: **C**

Explanation: Sense–Plan–Act is a conceptual cycle: perceive the world, decide/plan, then execute actions.

MCQ03 [Easy] Answer: **B**

Explanation: Reactive control maps current sensor readings directly to actions with minimal/no global model.

MCQ04 [Easy] Answer: **C**

Explanation: Hierarchical (deliberative) systems can be slow to react because planning depends on a world model and deliberation.

MCQ05 [Easy] Answer: **B**

Explanation: Hybrid systems keep a planner for global direction while a reactive layer handles real-time obstacles.

MCQ06 [Easy] Answer: **C**

Explanation: Hydraulic actuators use pressurized fluid to generate large forces/torques, common in heavy machinery.

MCQ07 [Easy] Answer: **B**

Explanation: Steppers move in discrete steps, enabling position control by counting pulses (often without feedback).

MCQ08 [Easy] Answer: **C**

Explanation: Servo systems use feedback (e.g., encoder/potentiometer) to reduce error between desired and actual position.

MCQ09 [Easy] Answer: **C**

Explanation: Rack-and-pinion converts gear rotation into linear translation of the rack.

MCQ10 [Easy] Answer: **B**

Explanation: LiDAR estimates distance by emitting laser pulses and measuring return time/phase.

MCQ11 [Easy] Answer: **B**

Explanation: Localization estimates where the robot is in its environment, often relative to a map or landmarks.

MCQ12 [Easy] Answer: **B**

Explanation: The Four Ds (Dirty, Dull, Dangerous, Difficult) identify tasks that are good candidates for robotics/automation.

MCQ13 [Easy] Answer: **B**

Explanation: Microcontrollers are used for real-time low-level tasks like reading sensors and controlling motors.

MCQ14 [Easy] Answer: **B**

Explanation: SBCs run an OS and support higher-level computation like vision, SLAM, and networking.

MCQ15 [Easy] Answer: **B**

Explanation: Pneumatic actuators use compressed air and are typically fast and clean for light/medium loads.

MCQ16 [Medium] Answer: **B**

Explanation: Worm drives can provide high reduction and torque with smooth motion, but typically lower efficiency.

MCQ17 [Easy] Answer: **B**

Explanation: The actuation/acting subsystem converts control commands into physical motion through actuators.

MCQ18 [Easy] Answer: **B**

Explanation: The First Law states a robot must not harm a human or allow a human to come to harm through inaction.

MCQ19 [Medium] Answer: **C**

Explanation: Reactive architectures generally avoid internal world models and long-term memory, relying on current sensing.

MCQ20 [Medium] Answer: **B**

Explanation: Hybrid systems retain global planning but handle sudden changes via reactive behaviors, improving robustness.

B) True/False Answers

TF01 [Easy] Answer: **True**

Explanation: Robotic systems are often described by Sense–Plan–Act and related subsystem breakdowns.

TF02 [Easy] Answer: **False**

Explanation: Reactive control avoids global models and maps sensor input directly to actions.

TF03 [Easy] Answer: **True**

Explanation: Hybrid architectures pair a deliberative planner with a reactive layer for real-time response.

TF04 [Easy] Answer: **False**

Explanation: Hydraulics are commonly used specifically because they provide high force/torque.

TF05 [Medium] Answer: **True**

Explanation: Because LiDAR uses its own emitted laser light, it does not depend on ambient illumination like cameras.

TF06 [Easy] Answer: **False**

Explanation: Microcontrollers are designed for low power embedded use; microprocessors typically consume more power.

TF07 [Medium] Answer: **True**

Explanation: Steppers can maintain holding torque at a commanded step position while powered.

TF08 [Easy] Answer: **False**

Explanation: They identify tasks that are dirty/dull/dangerous/difficult and are good candidates for robots.

TF09 [Easy] Answer: **True**

Explanation: Localization estimates where the robot is; planning computes how to reach a goal.

TF10 [Easy] Answer: **True**

Explanation: Servos use feedback to reduce error and maintain accurate position/speed/torque.

TF11 [Easy] Answer: **True**

Explanation: Compressed air actuation is often clean and offers quick response, though force is limited vs hydraulics.

TF12 [Easy] Answer: **False**

Explanation: Deliberative systems can be slow to respond because they depend on planning and model updates.

TF13 [Easy] Answer: **True**

Explanation: Autonomous Underwater Vehicles operate underwater without direct human control.

TF14 [Easy] Answer: **False**

Explanation: Electric actuators use electrical energy; oil/fluids are characteristic of hydraulic systems.

TF15 [Medium] Answer: **True**

Explanation: Reactive architectures emphasize direct sensor-to-action mappings via behaviors.

TF16 [Easy] Answer: **False**

Explanation: Belts are typically quieter; chains are noisier due to metal-on-metal contact.

TF17 [Easy] Answer: **True**

Explanation: PLCs are industrial-grade controllers valued for reliability and deterministic control.

TF18 [Easy] Answer: **False**

Explanation: Asimov’s laws explicitly prioritize human safety and obedience, then robot self-preservation.

TF19 [Easy] Answer: **True**

Explanation: A planner provides global direction; the reactive layer handles local real-time interactions.

TF20 [Easy] Answer: **True**

Explanation: Rack-and-pinion converts rotational input into linear output motion.

C) Descriptive Expected Answers

DES01 [Medium]

Expected answer: Hybrid; LiDAR as primary with optional camera; servo motor for controlled motion; belt drive for mop arm.

Explanation: Hybrid fits because the robot needs global route efficiency (planning) and fast reactions to people and sensor artifacts. LiDAR is generally more reliable than vision alone on glossy/reflective surfaces and in varying lighting; a camera can help with semantic detection. Servos provide smooth closed-loop control, and a belt drive is quieter and smoother for a cleaning arm than a chain, with sufficient precision for this task.

DES02 [Hard]

Expected answer: Electric actuators; Hybrid paradigm; Ultrasonic or LiDAR (with dust tradeoff) for obstacle detection.

Explanation: Electric actuation is typically the best balance of efficiency, controllability, and low maintenance for medium loads. A hybrid approach allows mission-level routing while still reacting quickly to people and pallets. Ultrasonic sensors can be robust in low light and relatively tolerant of dust; LiDAR can provide richer geometry but may suffer from dust scatter, so a careful answer should mention this tradeoff.

DES03 [Medium]

Expected answer: Hybrid (or Hierarchical for controlled setups); servo for smooth precise joint control; camera for object pose detection.

Explanation: Manipulation benefits from planned motion and safety checks, and a hybrid layer can also respond to nearby people. Servo motors provide closed-loop precision for joint angles and smooth motion, reducing the risk of damaging fragile objects. A camera enables identifying the object and estimating grasp pose; adding proximity sensing improves safety.

DES04 [Hard]

Expected answer: Hybrid; camera + IMU/GPS (optionally LiDAR/ultrasonic for obstacles); electric actuators for practicality and maintenance.

Explanation: Outdoor environments change quickly, so hybrid control gives goal-directed behavior while still reacting to terrain and obstacles. Row alignment often benefits from vision, while IMU and GPS support robustness when GPS degrades; LiDAR/ultrasonic can help with obstacle detection. Electric actuation is usually easier to power, control, and maintain for field robots than fluid systems.

DES05 [Medium]

Expected answer: Servo motors; PLC/industrial controller; gear transmission (e.g., planetary/harmonic depending on axis).

Explanation: Servo motors with feedback provide high precision and repeatability for industrial arms. PLCs/industrial controllers are standard for deterministic, reliable automation with robust I/O and safety integration. Gear transmissions provide stiffness and minimal slip compared to belts/chains, improving precision.